

INSECTICIDAL ACTIVITY OF TELESCOPE SNAILS (*Telescopium telescopium*) AGAINST RICE BLACK BUG (*Scotinophara coarctata*) UNDER IN VIVO & INVITRO CONDITIONS

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ABSTRACT

One of the most significant and invasive rice pests is the rice black bugs. Phytochemical analysis revealed that alkaloids, flavonoids and tannins are present in the telescope snail extract. The study had a total of three treatments with two trials replicated three times per trial and a Randomized Complete Block Design (RCBD) was enforced to ensure randomization. Screening of insecticidal activity of the telescope snail extract was done by spraying the extract all over the screenhouse and beaker. After days of observation, 100% Telescope Snail Extract showed the highest recorded mortality rate (M=9.333), followed by the 50% Telescope Snail Extract (M=8.333) while the positive control group (M=8.167). Using One-Way Analysis of Variance, the comparisons of means between the three treatments showed no significant difference in terms of the mortality rate of rice black bug (F=1.525, p<0.250). With the comparable means in terms of the insecticidal activity results, the researchers suggest and recommend that the telescope snail extract can be used as an alternative insecticide, suggestive of its expectation in terms of its insecticidal activity.

Keywords: Rice Black Bug, Telescope Snail Extract, Alkaloids, Flavonoids, Tannins

BACKGROUND OF THE STUDY

Rice black bugs are considered one of the major pests in rice (Catindig, n.d). They attack the crops by sucking out the plant sap at all stages of the plant's life through the stems and nodes near the water surface (Sumathi & Ramasubramanian, 2013).

Telescope Snails belong to the invertebrate family include phytochemicals like alkaloids, flavonoids & tannins, (Purwaningsih et al., 2019).

OBJECTIVES



To produce a cheap, organic, eco-friendly and effective pesticide to control the infestations of Rice Black Bugs in the country and other places as well.

REVIEW OF RELATED LITERATURE

A study by Latiza et al. (2022) investigated the effectiveness of Bignay (*Antidesma bunius*) leaf extract as an organic pesticide against Rice Black Bugs. The findings showed that plant-based extracts demonstrated pesticidal activity and may serve as a possible alternative to chemical pesticides, although certain concentrations affected rice crops. This supports the potential of developing organic pesticide solutions using natural materials.

METHODOLOGY

All photos were taken by the researchers.



BIBLIOGRAPHY

Colas, M. A., Gloria, R. C., Obafial, N. A. (2016). Insecticidal Activity of Alkaloids Extract of Acacia (*Samanea saman*) Bark Against Adult Rice Black Bug (*Scotinophara* sp.) Under Screen House Condition. Retrieved from: <https://abstracts.societyforscience.org/Home/PrintPdf/8873>

Latiza, S., Pasawilan, A., Gacoscos, G., Bangeles, D., Caas, C., Valdez, A., & Abusama, H. (2022, March 1). Bignay (*Antidesma Bunius*) Leaf Extract Stands as an Organic Pesticide against Rice Black Bugs (*Scotinophara Coarctata*). *ASEAN Journal of Science and Engineering*. Retrieved from <https://ejournal.upi.edu/index.php/AJSE/article/view/37781>

RESULTS AND DISCUSSION

	Ave	F-Stat	P-Value
Extract 100%	9.333	1.525	0.250
Extract 50%	8.333		
Metarhizium anisopliae (50mL)	8.167		

Table 1.

Test Sample	Phytochemicals (Qualitative)		
	Alkaloids	Flavonoids	Tannins
Telescopium Snails Extract (<i>Telescopium telescopium</i>)	+	+	+

Table 2.

Table 1 shows the result of the difference in the number of rice black bugs that were exposed to three different treatments (F=1.525; p<.250). The analysis, shown in Table 2, indicates that the *Telescopium* extract contains alkaloids, flavonoids and tannins.

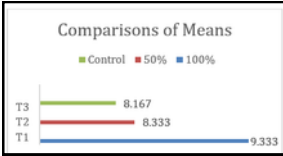


Table 3.

This table indicates that the 100% extract showed the highest mortality rate of RBB (M=9.333), followed by the 50% extract (M=8.333). and positive control group showed the least mortality rate of RBB (M=6.833).

CONCLUSION

The recorded result under both in vivo and in vitro conditions showed no significant difference between the three treatments (F=1.525; p<.250). With the comparable means in the results, the telescope snail extract can be used as an alternative insecticide.

RECOMMENDATIONS

- Future studies & researchers should:
- Explore other snail's effectiveness in killing rice black bugs.
 - Explore different insects that infests the agricultural sector.
 - Increase the number of rice black bugs being tested per trial and increase the number of trials for more reliable results.